

UCT203: DATA STRUCTURES AND ALGORITHMS

L	T	P	Cr
3	0	2	4.0

Course Objectives: To become familiar with different types of data structures, algorithms and their applications.

Basic Terminologies and Introduction to Algorithm & Data Organization: Algorithm specification, Recursion, Performance analysis, Asymptotic Notation - The Big-O, Omega and Theta notation, Refinement of Coding - Time-Space Trade Off, Testing, Data Abstraction

Linear Data Structure: Array, Stack, Queue, Linked-list and its types, Various Representations, Operations & Applications of Linear Data Structures

Non-linear Data Structure: Binary Tree, Binary Search Tree, Balancing of Search Trees – AVL Trees, Operations & Applications of Non-Linear Data Structures

Searching and Sorting on Various Data Structures: Divide-and-Conquer – Review, Linear Search, Binary Search, Insertion Sort, Merge Sort, Quick Sort, Heapsort, Radix sort, bucket sort, Introduction to Hashing

Graph: Basic Terminologies and Representations, Graph search and traversal algorithms and complexity analysis.

Laboratory work:

Implementation of Arrays, Recursion, Stacks, Queues, Lists, trees, graphs, Sorting techniques, Searching techniques.

Course Learning Outcomes (CLOs)/ Course Objectives (COs):

On completion of this course, the students will be able to:

1. Apply basic concepts of asymptotic notation in analysis of algorithm.
2. Understand and implement various linear data structures: Arrays, Stack, Queue and Link list.
3. Implement various searching and sorting techniques based on characteristics of application.
4. Understand and Implement various Non linear data structures: Trees and Graphs.

Text Books:

1. Introduction to Algorithms, Thomas, H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, The MIT Press.
2. Fundamentals of Data Structures, E. Horowitz, S. Sahni, S. A-Freed, Universities Press.
3. Data Structures and Algorithms, A. V. Aho, J. E. Hopcroft, J. D. Ullman, Pearson.

Reference books:

1. The Art of Computer Programming: Volume 1: Fundamental Algorithms, Donald E. Knuth.
2. Open Data Structures: An Introduction (Open Paths to Enriched Learning), (Thirty First Edition), Pat Morin, UBC Press.
3. Micheal T. Goodrich and Roberto Tamassia: Algorithm Design: Foundations, Analysis and Internet examples (John Wiley & Sons, Inc., 2002).